

Rational horizon and uniform dyadic-root trace margin boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope firewall

R-116 reduced a same-root visit family to one endpoint square-minus-trace quotient and isolated a rational-homogeneous recession problem. It also classified the R-082 active finite-Fourier nulls, but left open a cutoff-uniform near-null metric estimate. The present result takes the stronger route: it proves a strict trace margin in every standardized canonical dyadic-root direction, so no null-set metric estimate is needed for the bare root's tail integrability.

What is proved. A full-horizon theorem extends the R-116 polynomial recession classifier to locally Lipschitz asymptotically quadratic maps, including the rational R-082 row. The positive floor converges uniformly to its degree-two horizon at rate $O(t^{-1})$. For the homogeneous Pauli-Fierz frame, the exact joint trace constant is

$$\kappa_* = \frac{411}{2000P}, \quad P = 4 + 10^{-12}.$$

For the canonical full-production dyadic shell, every six-real standardized unit direction obeys

$$\frac{10}{9}\tau_N^{\text{rec}}(v) < \frac{3}{40} < \frac{2}{25}, \quad \frac{20}{9}\tau_N^{\text{rec}}(v) < \frac{3}{20} < \frac{4}{25}.$$

Thus the frozen bare-root normalizers at both tilts have strictly positive leading Gaussian precision. This covers all directions and hence all orthogonally projected null directions. A same-shell phase modulation gives an exact fixed-cutoff failure of any local Lipschitz metric error bound.

What is not proved. The trace theorem concerns the degree-two recession of one canonical shell with its matching covariance, fixed independent heat, and fixed orthogonal output projection. It is not a raw finite-amplitude heat estimate, does not cover an old low carrier paired with cumulative high covariance, and does not itself identify the complete R-107/R-116 progressive or revisit packet. In particular it supplies finiteness, not a cutoff-summable sequence of log-normalizer constants. The complete owner-preserving embedding, global one-use source/sextic aggregation, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, removals, interacting measure, Sector-A closure, and tier promotion remain open. A13 stays T4.

All production constants below come from the hash-pinned A1 production manifest. The scalar quadratic symbol is used only as a safe lower bound on the full 3×3 internal symbol; the nonnegative family and locking mass matrix can only improve the estimates.

2 A full-horizon tail theorem for rational maps

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be locally Lipschitz with at most quadratic growth, let $\tau : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous with at most quadratic growth, and assume the uniform spherical limits, in the explicit sense

$$\sup_{|v|=1} \left\| \frac{F(rv)}{r^2} - Q_\infty(v) \right\| + \sup_{|v|=1} \left| \frac{\tau(rv)}{r^2} - \tau_\infty(v) \right| \rightarrow 0. \quad (2.1)$$

The maps on the right are continuous. Put

$$\mathcal{Z}_\infty = \{v \in S^{n-1} : Q_\infty(v) = 0\}. \quad (2.2)$$

For $v \in \mathcal{Z}_\infty$, define the full horizon tangent image

$$\mathcal{T}_F(v) = \left\{ z : \begin{array}{l} r_j \rightarrow \infty, \ v_j \rightarrow v, \ |v_j| = 1, \\ F(r_j v_j)/r_j \rightarrow z \end{array} \right\}. \quad (2.3)$$

It can be empty and need not equal the range of a first derivative. Given $b \in \mathbb{R}^m$, a linear map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and $q > 0$, set

$$U_q(x) = \frac{1}{2} \left\{ |x|^2 - q\tau(x) + q\|b + Ax + F(x)\|^2 \right\}. \quad (2.4)$$

Theorem 2.1 (full rational-horizon classifier)

With the convention $\inf \emptyset = +\infty$,

$$\liminf_{|x| \rightarrow \infty} \frac{2U_q(x)}{|x|^2} = \lambda_q := \inf_{\substack{v \in \mathcal{Z}_\infty \\ z \in \mathcal{T}_F(v)}} \{1 - q\tau_\infty(v) + q\|Av + z\|^2\}. \quad (2.5)$$

If $\lambda_q > 0$, then e^{-U_q} is integrable. If $\lambda_q < 0$ and there are $C, M, R_0 < \infty$ such that

$$\text{Lip}(U_q; \{R/2 \leq |x| \leq 2R\}) \leq CR^M \quad (R \geq R_0), \quad (2.6)$$

then e^{-U_q} is not integrable. The equality case $\lambda_q = 0$ is not decided by recession alone.

Proof. Write $x_j = r_j v_j$. If $2U_q(x_j)/r_j^2$ is bounded above, positivity of the last square in (2.4) makes $F(r_j v_j)/r_j$ bounded. After a subsequence, $v_j \rightarrow v$ and $F(r_j v_j)/r_j \rightarrow z$. Uniformity in (2.1) then forces $Q_\infty(v) = 0$, while $b/r_j \rightarrow 0$, and the subsequential limit is the quantity in braces in (2.5). Conversely, every $z \in \mathcal{T}_F(v)$ is realized by its defining sequence and gives the reverse inequality. This proves (2.5).

If $\lambda_q > 0$, (2.5) gives $U_q(x) \geq c|x|^2$ outside a ball. If $\lambda_q < 0$, a realizing sequence has $U_q(x_j) \leq -c|x_j|^2$. A polynomial annular Lipschitz bound preserves this inequality, with a smaller c , on balls of radius comparable to $\min\{r_j, r_j^{2-M}\}$. Passing to a separated subsequence, their volumes are only polynomially small while their exponential masses grow like $e^{cr_j^2}$, so the sum diverges. $\square$

This theorem is deliberately formulated with $\mathcal{T}_F(v)$ rather than $\text{Ran } DF(v)$. R-116 already showed that singular parabolic approaches can create horizon tangents outside the derivative range.

3 The R-082 positive-floor horizon

Write a state and tangent as

$$z = (u_1, u_2, \chi), \quad y = (v_1, v_2, w), \quad (3.1)$$

and put

$$r = |u|^2, \quad \rho = r + |\chi|^2, \quad dr = 2\Re(u^* v), \quad d\rho = 2\Re(u^* v + \bar{\chi} w), \quad h = u_1 v_2 - u_2 v_1. \quad (3.2)$$

For $e > 0$, the R-082 four-coordinate map is

$$\Xi_e(z, y) = \begin{pmatrix} \sqrt{c_0} dr \\ \sqrt{c_1} \{dr - \alpha r(\rho + e)^{-1} d\rho\} \\ 2\sqrt{c_0 + c_1} \Re h \\ 2\sqrt{c_0 + c_1} \Im h \end{pmatrix}, \quad (3.3)$$

where

$$c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}, \quad \alpha = \frac{5}{9}. \quad (3.4)$$

At $e = 0$, the ratio $\beta = r/\rho$ has no continuous value at $\rho = 0$. Nevertheless the product $g_\beta = ((1 - \alpha\beta)u, -\alpha\beta\chi)$ and the complete map Ξ_0 extend continuously by zero, because $|g_\beta|^2 \leq \rho$.

Proposition 3.1 (uniform positive-floor horizon)

For $t > 0$,

$$\Xi_e(tz, ty) = t^2 \Xi_{e/t^2}(z, y), \quad (3.5)$$

and, uniformly in z ,

$$\|t^{-2} \Xi_e(tz, ty) - \Xi_0(z, y)\| \leq \sqrt{c_1} \alpha \frac{\sqrt{e}}{t} |y|. \quad (3.6)$$

Proof. Only the second row changes. With $\delta = e/t^2$, its absolute difference is at most

$$\sqrt{c_1} \alpha \frac{r\delta}{\rho(\rho + \delta)} |d\rho| \leq 2\sqrt{c_1} \alpha \frac{\delta\sqrt{\rho}}{\rho + \delta} |y|. \quad (3.7)$$

The square of the normalized scalar factor is

$$\frac{1}{\delta} \left(\frac{2\delta\sqrt{\rho}}{\rho + \delta} \right)^2 = \frac{4x}{(1+x)^2} \leq 1, \quad x = \frac{\rho}{\delta}, \quad (3.8)$$

because its defect from one is $(x-1)^2/(x+1)^2$. $\square$

The bound is uniform in z for the displayed tangent y . After inserting a derivative $y = Dz$, its spherical $O(t^{-1})$ constant can depend on the finite cutoff. Proposition 3.1 proves the horizon separately at every cutoff; it is not a cutoff-uniform horizon-regularity estimate.

For fixed target heat R independent of the fresh root, (3.6) and the linear frame bound below give dominated convergence:

$$\lim_{t \rightarrow \infty} t^{-2} P_\Sigma B_e(tz) = B_\infty(z). \quad (3.9)$$

This is a recession statement. The raw finite-amplitude estimate $P_\Sigma B_e(z) \lesssim |z|^2$ is false at $z = 0$ whenever the heat has positive variance.

4 Sharp homogeneous frame trace

Let Y_i be a proper circular complex Gaussian derivative with covariance $D_i = \mathbb{E}[Y_i Y_i^*]$, and write $d_i = \|D_i\|_{\text{op}}$. Put $\beta = r/\rho$ and

$$g_\beta = ((1 - \alpha\beta)u, -\alpha\beta\chi). \quad (4.1)$$

Then

$$\frac{|g_\beta|^2}{\rho} = f(\beta) := \beta - \alpha(2 - \alpha)\beta^2 = \beta - \frac{65}{81}\beta^2. \quad (4.2)$$

The separate rational row has maximum $81/260$, but optimizing all four rows jointly is sharper.

Theorem 4.1 (sharp homogeneous frame trace constant)

For every z and each spatial direction,

$$\mathbb{E}\|\Xi_0(z, Y_i)\|^2 \leq d_i \rho K(\beta), \quad (4.3)$$

where

$$K(\beta) = 6(c_0 + c_1)\beta - 2c_1 \frac{65}{81} \beta^2. \quad (4.4)$$

The function is increasing on $[0, 1]$, and hence

$$\boxed{\mathbb{E}\|\Xi_0(z, Y_i)\|^2 \leq \kappa_* d_i \rho, \quad \kappa_* = \frac{411}{2000P}.} \quad (4.5)$$

The constant is attained when $\chi = 0$ and D_i is scalar on the active doublet.

Proof. Proper complex covariance gives

$$\mathbb{E}(dr)^2 = 2u^* D_i u \leq 2d_i r, \quad \mathbb{E}\{2\Re(g_\beta^* Y_i)\}^2 \leq 2d_i |g_\beta|^2. \quad (4.6)$$

Also

$$\mathbb{E}|u_1(Y_i)_2 - u_2(Y_i)_1|^2 \leq d_i r. \quad (4.7)$$

After multiplying by the four row weights, the sum is (4.3)–(4.4). Moreover

$$K'(1) = 6(c_0 + c_1) - 4c_1 \frac{65}{81} > 0, \quad (4.8)$$

and K' decreases linearly, so K is increasing. Finally

$$K(1) = 6c_0 + \left(4 + \frac{32}{81}\right) c_1 = \frac{411}{2000P}. \quad (4.9)$$

The covariance convention is $\Gamma_{\mathbb{R}} = \text{realify}(D_i)/2$; (4.6) already contains this factor. There is no additional Pauli, complex, or volume multiplier. $\square$

5 Canonical dyadic-root uniform trace margin

Let $L = 16$, $V = L^3 = 4096$, $a = 2\pi/L = \pi/8$, and

$$\Lambda_N = \{n \in \mathbb{Z}^3 : N/2 < \|n\|_\infty \leq N\}, \quad N = 1, 2, 4, \dots \quad (5.1)$$

The full production quadratic symbol is

$$A(n) = p(|k_n|^2) \mathbf{I} + M, \quad p(s) = r_0 + Zs + Ys^2, \quad k_n = an, \quad (5.2)$$

where $M \succeq 0$, $r_0 = 0.4740336473$, $Z = -0.9252754126$, and $Y = 1$. Define safe scalar quantities

$$m_N = \min_{n \in \Lambda_N} p(|k_n|^2), \quad (5.3)$$

$$d_N = \frac{2}{V} \sum_{n \in \Lambda_N} \frac{k_{n,i}^2}{p(|k_n|^2)}. \quad (5.4)$$

Cube symmetry makes (5.4) independent of i , and operator monotonicity gives $D_{N,i}^{\mathbb{C}} \preceq d_N \mathbf{I}$.

For a unit vector v in the six-real standardized shell coordinate, let z_v be its physical field direction. The deliberately conservative Parseval convention is

$$\int_{\mathbb{T}^3} \rho(z_v) \leq \frac{2}{m_N}. \quad (5.5)$$

The sharp convention can remove the factor two; retaining it makes the statement insensitive to the equivalent complex whitening normalization.

Theorem 5.1 (uniform dyadic-root recession trace margin)

Let $\tau_N^{\text{rec}}(v)$ be the sum over the three spatial derivatives of the degree-two recession trace of the frozen canonical R-082 root, possibly followed by a fixed orthogonal output projection. Then, for every dyadic $N \geq 1$ and every standardized unit direction v ,

$$\tau_N^{\text{rec}}(v) \leq 6\kappa_* \frac{d_N}{m_N}. \quad (5.6)$$

At the two tilts required by R-116,

$$\frac{10}{9}\tau_N^{\text{rec}}(v) < \frac{3}{40} < \frac{2}{25}, \quad \frac{20}{9}\tau_N^{\text{rec}}(v) < \frac{3}{20} < \frac{4}{25}. \quad (5.7)$$

Consequently the leading Gaussian precisions are greater than $37/40$ and $17/20$, respectively.

Proof. Sum (4.5) over three axes and use (5.5), obtaining (5.6). Since $P > 4$,

$$\frac{10}{9}6\kappa_* < \frac{137}{400}. \quad (5.8)$$

The exact rational finite-shell enclosure, using the independently certified $3.14159 < \pi < 3.14160$, is

N	$\#\Lambda_N$	m_N lower	$(10/9)\tau_N^{\text{rec}}$ upper
1	26	0.260000000009474940	0.006407024335204
2	98	0.283781194978547397	0.032596740075668
4	604	1.11613025030910356	0.013780340467900

(5.9)

No floating-point comparison is used: the executable minimizes the convex rational polynomial on each rational π^2 interval and performs every sum as a fraction.

For $N \geq 8$, every shell frequency satisfies $|k_n|^2 > a^2 N^2/4 \geq 2|Z|/Y$, and therefore

$$p(|k_n|^2) \geq \frac{Y}{2}|k_n|^4. \quad (5.10)$$

Also $(2N+1)^3 \leq (17N/8)^3$. Cube symmetry gives

$$d_N \leq \frac{4913}{96Va^2}N, \quad m_N \geq \frac{a^4 N^4}{32}, \quad (5.11)$$

and hence

$$\frac{d_N}{m_N} \leq \frac{4913}{3Va^6 N^3} \leq \frac{4913}{24\pi^6} \left(\frac{8}{N}\right)^3. \quad (5.12)$$

At $N = 8$, (5.8), (5.12), and $\pi > 3.14159$ give the exact rational upper bound

$$\frac{10}{9}\tau_N^{\text{rec}}(v) \leq 0.0729287992478\dots < \frac{3}{40}. \quad (5.13)$$

The right side decreases like N^{-3} . This proves (5.7). $\square$

Corollary 5.2 (frozen bare-root normalizer existence)

For the canonical shell map with fixed independent heat and a fixed orthogonal output projection, positivity of the output square gives directly

$$U_q(x) \geq \frac{1}{2}\{|x|^2 - q\tau_N(x)\}. \quad (5.14)$$

The uniform trace horizon and (5.7) therefore give a positive quadratic lower bound outside a ball at $q = 10/9$ and $q = 20/9$. Both fixed-cutoff conditional normalizers exist. This direct argument also covers the $L^2(\mu_\Sigma; \mathbb{R}^4)$ heat-lifted output without invoking finite-dimensional compactness in Theorem 2.1. It is an existence and leading-precision statement; it does not bound the log-normalizer by a summable shell sequence.

6 The metric route fails even at fixed cutoff

Set $\kappa = 2\pi/L$ and work in the shell Λ_8 . With a fixed active polarization e_1 and a real amplitude $p > 0$, define

$$u_t(x) = pe^{6i\kappa x_1}\{1 + it \cos(\kappa x_1)\}e_1, \quad \chi_t = 0. \quad (6.1)$$

Its Fourier indices 5, 6, 7 all belong to the same shell. Let $\mathcal{Z}_8$ be the exact zero set of the full three-direction homogeneous current in this finite shell. Directly,

$$r_t = p^2\{1 + t^2 \cos^2(\kappa x_1)\}, \quad dr_t = -p^2\kappa t^2 \sin(2\kappa x_1), \quad h_t = 0. \quad (6.2)$$

Thus

$$\|\Xi_0(u_t, \partial u_t)\|_{L^2}^2 = \frac{Vp^4\kappa^2 t^4}{2} \left\{ c_0 + c_1 \left(\frac{4}{9}\right)^2 \right\}, \quad (6.3)$$

where the bracket equals $9/(500P) > 0$.

R-116 says that active finite-Fourier nulls are plane waves (with an independent singlet plane wave allowed). Fourier orthogonality therefore gives, for small t ,

$$\text{dist}_{L^2}(u_t, \mathcal{Z}_8)^2 = \frac{Vp^2 t^2}{2}. \quad (6.4)$$

The pure-singlet branch remains a fixed positive distance away. Every norm on this fixed finite-dimensional shell is equivalent, so

$$\frac{\text{dist}(u_t, \mathcal{Z}_8)}{\|\Xi_0(u_t, \partial u_t)\|} \asymp \frac{|t|}{t^2} \rightarrow \infty. \quad (6.5)$$

The standardized unit-sphere path $\hat{u}_t = u_t/\sqrt{1 + t^2/2}$ differs from u_t only by a radial $1 + O(t^2)$ factor. Homogeneity preserves the $O(|t|)$ distance versus $O(t^2)$ current separation, so the same conclusion holds in the R-116 standardized sphere metric.

Proposition 6.1 (fixed-shell metric-regularity no-go)

No local Lipschitz error bound

$$\text{dist}(u, \mathcal{Z}_8) \leq C\|\Xi_0(u, \partial u)\| \quad (6.6)$$

holds near the active plane wave in (6.1). Equivalently, for $\zeta = ipe^{6i\kappa x_1} \cos(\kappa x_1)e_1$,

$$DQ_\infty(v)\zeta = 0, \quad Q_\infty(\zeta) \neq 0. \quad (6.7)$$

This retires the proposed metric route. It is not a counterexample to the normalizer: Theorem 5.1 bypasses the null geometry with an all-direction trace margin.

7 Owner-preserving successor and the two remaining burdens

R-104 and R-116 give, for fixed reveal field, target heat, covariance, and contraction-closed projection, the exact same-root quotient

$$P_{j,C} = \frac{1}{2}(\|x_*\|^2 - \|x_0\|^2) - \frac{1}{2}(t_* - t_0). \quad (7.1)$$

Changing target heat requires the value-plus-heat edge or its compensator; it is not licensed by (7.1). The complete R-104 expectation ledger is

$$\langle P^{\text{comp}} \rangle = \langle B_{\text{low}} \rangle + L_{\text{low}} + \langle C_{\text{far}} \rangle + \langle \Delta W_{L\Sigma} \rangle + \langle \Delta W_R \rangle + \langle P_{\text{paid}} \rangle. \quad (7.2)$$

The owner multiplicities after quotienting are:

Owner	Multiplicity
conditional low object	1
complete low object	1
complete linear module (heat/trace/forest internal)	1
future raw Wick (fixed-low raw Wick remains low)	1
unshifted rational coordinate	1
shifted rational coordinate with terminal square	1
paid difference	1
Cartan FAR owner in REG only	1
q/r , Schur, matrix, and R-063 forest payments beyond their owner	0
obsolete base cubic and duplicate R-086 rows	0

(7.3)

Owner-specific partial-Wick bookkeeping is essential as a diagnostic, not as a replacement owner: full Wick would delete the physical ΣQ term. Fresh first chaos belongs inside the frozen whole-output determinant. If a random- W score is separated, R-116 requires both q and $2q$ normalizers; Theorem 5.1 now supplies the bare canonical recession margin at both tilts.

The remaining coherent endpoint term has the form

$$R_{\text{coh}}(\xi) = \Re\langle b + A\xi, R(\xi) \rangle + \frac{1}{2}\|R(\xi)\|^2 - \frac{1}{2}\delta\tau(\xi). \quad (7.4)$$

One sufficient next route would identify the degree-two recession of the entire owner-preserving conditional packet with a fixed contraction of the canonical root covered by Theorem 5.1. Such an identification is not claimed to be unique and may fail for revisit/future response. The actual objective is a complete adapted signed-packet conditional normalizer retaining endpoint heat-trace differences, mixed predictable baselines, and the R-063 forest exactly once.

Here is the precise conditional schema that would be sufficient after such a local theorem. Let $(\mathcal{F}_j)_{j=0}^J$ be the root filtration and put

$$X_j = P_j^{\text{comp}} + \eta_j \mathcal{E}_j + \zeta_j \mathcal{Y}_j. \quad (7.5)$$

Require X_j to be $\mathcal{F}_j$ -measurable, every earlier partial sum to be $\mathcal{F}_{j-1}$ -measurable, and $\eta_j, \zeta_j \geq 0$ to be deterministic or predictable. Let c_j be fixed in advance, either deterministic or $\mathcal{F}_{j-1}$ -measurable, with $\sum_j c_j \leq C$ pathwise. If

$$\log \mathbb{E} \left[e^{-qX_j} \mid \mathcal{F}_{j-1} \right] \leq c_j, \quad q = \frac{10}{9}, \quad (7.6)$$

and the local charges obey the separate pathwise owner domination

$$\sum_j \eta_j \mathcal{E}_j \leq \frac{9}{20} \sum_j \|h_j\|^2, \quad \sum_j \zeta_j \mathcal{Y}_j \leq \frac{3}{20} \|Z_K\|_6^6, \quad (7.7)$$

then conditional iteration gives $\mathbb{E} e^{-q \sum_j X_j} \leq e^C$. The two global budgets become $1/2$ and $1/6$ after multiplication by q . Coefficient sums alone do not imply (7.7) for different parent fields. A common terminal sextic is future measurable and therefore must either be localized predictably or paid once outside the one-step tower; this is part of the open gate.

A smallest direct signed target remains

$$\sum_j \mathbb{E} P_j^{\text{comp}} \geq -\eta \mathbb{E} \sum_j \|h_j\|^2 - \zeta \mathbb{E} \|Z_K\|_6^6 - C, \quad 0 \leq \eta < \frac{9}{20}, \quad 0 \leq \zeta < \frac{3}{20}, \quad (7.8)$$

together with the adapted exponential upgrade (7.5)–(7.7).

Uniform strict precision alone does not make $\sum_j c_j$ finite. A trace-only determinant bound can cost a positive constant on every shell, and R-110 already records the positive-density bounded-multiplicity nonsummability boundary. The two remaining Sector-A burdens are therefore the legal complete adapted packet normalizer and, after it, quantitative summability with signed one-use owner aggregation. Bare null-cone integrability is no longer the obstruction.

8 Proof-route evidence map

Route	Verdict	Research-reusable reason
polynomial-only recession	superseded	Theorem 2.1 uses the full horizon tangent image and covers the rational row
floor treated as exactly homogeneous	repaired	(3.5)–(3.8) give the state-uniform fixed-cutoff $O(t^{-1})$ horizon error
null-cone metric stability	failed	the same-shell phase fixture has distance $O(t)$ and current $O(t^2)$
canonical same-shell all-direction trace	accepted	the sharp frame constant and scalar-symbol shell bounds give (5.7)
pure scalar authority	rejected	the proof uses the full A1 production manifest and only discards a PSD internal mass
raw heat trace bound	rejected	fixed heat disappears only after degree-two recession scaling
state-dependent compression	excluded	only fixed orthogonal output projections contract the trace
cumulative covariance on old carrier	excluded	m_N and D_N must come from the same fresh dyadic shell
same-root visit-by-visit payment	superseded	the exact object is the endpoint quotient (7.1)
trace margin implies summability	failed	strict per-shell precision does not imply $\sum_j c_j < \infty$
owner-preserving supermartingale	open schema	prove measurability, the local conditional bound, and owner domination (7.5)–(7.7)

The primary executable uses exact rational interval arithmetic, a tight Machin enclosure of π , termwise shell enumeration, and the joint frame constant. The non-importing verifier instead uses a coarser $157/50 < \pi < 22/7$ enclosure, radius/axis multiplicities, and the six-real Frobenius identity. They agree on every theorem-level inequality.

9 Devil’s-advocate review

1. **Objection:** “The 81/260 rational-row maximum is the joint frame constant.” **VALID WITH MITIGATION.** Maximizing rows separately is safe but not sharp. Joint optimization gives $\kappa_* = 411/(2000P)$ in (4.5).
2. **Objection:** “A complex-to-real factor two is missing.” **DISMISSED.** The covariance is $\Gamma_{\mathbb{R}} = \text{realify}(D)/2$; equations (4.6)–(4.7), three axes, and the conservative factor two in (5.5) account for every factor.
3. **Objection:** “The internal production mass can lower the symbol.” **DISMISSED.** It is a sum of a nonnegative family diagonal and a nonnegative locking projection. Dropping it makes both (5.3) and (5.4) worse in the safe direction.

4. **Objection: “The finite-shell scan is floating-point evidence.” DISMISSED.** Both executables first prove a rational enclosure of π , minimize the convex symbol on rational intervals, and compare exact fractions. Printed decimals are display values only.
5. **Objection: “The positive floor invalidates the homogeneous row maximum.” VALID WITH MITIGATION.** The maximum is used only for the degree-two horizon. Proposition 3.1 proves uniform convergence; no finite-amplitude floor claim is made.
6. **Objection: “Fixed target heat obeys the same raw trace estimate.” UPHELD AGAINST THAT WORDING.** At zero state the heat lift can be positive. Only its radial recession is (3.9).
7. **Objection: “The null classification yields a metric error bound.” UPHELD AGAINST THAT ROUTE.** Proposition 6.1 is an explicit same-shell counterexample. The accepted trace route does not use metric regularity.
8. **Objection: “All-direction trace control automatically covers arbitrary clusters.” VALID WITH MITIGATION.** Fixed orthogonal projections contract the Hilbert norm. Nonorthogonal, duplicated, or state-dependent output selection is outside Theorem 5.1.
9. **Objection: “The doubled-tilt margin closes the complete one-fresh-root packet.” UPHELD AGAINST THAT CLAIM.** The complete adapted signed-packet conditional normalizer in Section 7 has not been proved; a fixed contraction is only one possible route.
10. **Objection: “Small coefficient sums make the one-use tower automatic.” UPHELD AGAINST THAT CLAIM.** Different parent fields can share future information. The measurability and pathwise owner domination in (7.5)–(7.7) are separate load-bearing hypotheses.
11. **Objection: “Uniform normalizer existence is the needed one-use estimate.” UPHELD AGAINST THAT CLAIM.** Sector A needs a summable c_j ledger and one global source/sextic spend as in (7.5)–(7.7). Strict tail precision alone is insufficient.
12. **Objection: “The horizon tangent proof automatically extends to an infinite-dimensional output.” UPHELD AGAINST THAT WORDING.** Theorem 2.1 uses compactness in $\mathbb{R}^m$. Corollary 5.2 is instead proved directly from the nonnegative output square and the uniform trace horizon, so no Hilbert-output compactness is assumed.

10 Result footer

Result ID: A13-CLASSII-RATIONAL-HORIZON-UNIFORM-ROOT-TRACE-MARGIN-BOUNDARY

Ledger: R-117.

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.

Precise statement: Theorems 2.1, 4.1, and 5.1; Propositions 3.1 and 6.1; Corollary 5.2; owner-preserving successor boundary (7.1)–(7.8).

Scope: canonical same-shell full-production dyadic root; degree-two recession; fixed independent heat; fixed orthogonal output projection; fixed-cutoff normalizer existence.

Dependencies: R-063, R-082, R-104, R-107, R-110, R-116 and the hash-pinned A1 production manifest.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_rational_horizon_uniform_root_trace_margin_boundary_verify.py
```

Expected output: Integrated R-117 PASS; primary 42/42 and independent 36/36 children PASS, with all authority, source, note, PDF, and manifest

pins valid.
 Falsification gate: any failed horizon, frame-factor, shell-interval,
 high-frequency, phase-modulation, child/source/note/PDF/manifest assertion;
 any use beyond the stated same-shell recession scope.
 Tier before / after: T4 / T4.
 Open gates: A13-CLASSII-FULL-PROGRESSIVE-REVISIT-EXTENSION;
 A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE.
 No-overclaim statement: No complete progressive/revisit packet estimate,
 cutoff-summable log-normalizer, one-use source/sextic aggregation,
 OVERLAP_src, Nelson estimate, removal limit, interacting measure,
 Sector-A closure, or tier promotion is established.
 Next required action: prove a legal complete adapted conditional-packet
 normalizer, potentially through a fixed canonical contraction; then prove
 a summable c_j bound and aggregate source energy and terminal sextic once.